

Exceptions

1. Write a program to generate Arithmetic Exception without exception handling

Answer:

```
public class Demo {  
    public static void main(String[] args) {  
        int a = 10, b = 0;  
        System.out.println(a / b);  
    }  
}
```

2. Handle the Arithmetic exception using try-catch block

Answer:

```
public class Demo {  
    public static void main(String[] args) {  
        try {  
            int a = 10, b = 0;  
            System.out.println(a / b);  
        } catch (ArithmeticException e) {  
            System.out.println("Cannot divide by zero.");  
        }  
    }  
}
```

3. Write a method which throws exception, Call that method in main class without try block

Answer:

```
public class Demo {  
    static void test() throws Exception {  
        throw new Exception("Exception from method");  
    }  
    public static void main(String[] args) throws Exception {  
        test();  
    }  
}
```

4. Write a program with multiple catch blocks

Answer:

```
public class Demo {  
    public static void main(String[] args) {  
        try {  
            int[] a = new int[2];  
            a[5] = 10;  
        } catch (ArithmeticException e) {  
            System.out.println("Arithmetic Exception");  
        } catch (ArrayIndexOutOfBoundsException e) {  
            System.out.println("Array Index Exception");  
        } catch (Exception e) {  
            System.out.println("General Exception");  
        }  
    }  
}
```

5. Write a program to throw exception with your own message

Answer:

```
public class Demo {  
    public static void main(String[] args) {  
        throw new ArithmeticException("Division by zero is not  
allowed.");  
    }  
}
```

6. Write a program to create your own exception

Answer:

```

class MyException extends Exception {
MyException(String msg) {
    super(msg);
}
}

public class Demo {
    public static void main(String[] args) {
        try {
            throw new MyException("My Custom Exception");
        } catch (MyException e) {
            System.out.println(e.getMessage());
        }
    }
}

```

7. Write a program with finally block

Answer:

```

public class Demo {
    public static void main(String[] args) {
        try {
            int a = 10 / 0;
        } catch (ArithmeticException e) {
            System.out.println("Exception Handled");
        } finally {
            System.out.println("Finally Block Executed");
        }
    }
}

```

8. Write a program to generate Arithmetic Exception

Answer:

```
public class Demo {  
    public static void main(String[] args) {  
        int a = 10 / 0;  
    }  
}
```

9. Write a program to generate ArrayIndexOutOfBoundsException

Answer:

```
public class Demo {  
    public static void main(String[] args) {  
        int[] a = {1,2,3};  
        System.out.println(a[5]);  
    }  
}
```

10. Write a program to generate ClassNotFoundException

Answer:

```
public class Demo {  
    public static void main(String[] args) throws  
        ClassNotFoundException {  
        Class.forName("Test");  
    }  
}
```

11. Write a program to generate FileNotFoundException

Answer:

```
import java.io.*;

public class Demo {

    public static void main(String[] args) throws
FileNotFoundException {

        FileInputStream fis = new FileInputStream("abc.txt");

    }

}
```

12. Write a program to generate IOException

Answer:

```
import java.io.*;

public class Demo {

    public static void main(String[] args) throws IOException
{

        FileReader fr = new FileReader("abc.txt");

        fr.read();

        fr.close();

    }

}
```

13. Write a program to generate NoSuchFieldException

Answer:

```
public class Demo {

    int id;

    public static void main(String[] args) throws Exception {

        Demo.class.getField("name");

    }

}
```

14. Write a program to generate NoSuchMethodException

Answer:

```
public class Demo {  
    public static void main(String[] args) throws Exception {  
        Demo.class.getMethod("display");  
    }  
}
```

15. Write a program to generate NullPointerException

Answer:

```
public class Demo {  
    public static void main(String[] args) {  
        String s = null;  
        System.out.println(s.length());  
    }  
}
```

16. Write a program to generate NumberFormatException

Answer:

```
public class Demo {  
    public static void main(String[] args) {  
        int n = Integer.parseInt("ABC");  
    }  
}
```

17. Write a program to generate StringIndexOutOfBoundsException

Answer:

```
public class Demo {  
    public static void main(String[] args) {  
        String s = "Java";  
        System.out.println(s.charAt(10));  
    }  
}
```

18. Write a program to generate SQLException

Answer:

```
import java.sql.*;  
public class Demo {  
    public static void main(String[] args) throws SQLException  
    {  
        DriverManager.getConnection(  
            "jdbc:mysql://localhost:3306/test",  
            "wronguser",  
            "wrongpassword"  
        );  
    }  
}
```